



Airway Management & Ventilation

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined HandTevy
- Children compensate well initially but decompensate quickly with little warning.
- Most pediatric cardiac arrests are due to respiratory compromise.

Signs & Symptoms:

- Reason for tracheostomy placement
- Time since tracheostomy placement
- Bleeding
- Tracheal trauma
- Respiratory distress
- Swelling/edema
- Closing of tracheostomy

Differential:

- Airway obstruction
- Blocked tube
- Mucosal or tracheal injury
- Tracheo-innominate fistula
- Pneumonia
- Equipment malfunction
- COPD/Asthma
- Pulmonary Edema

Clinical Management Options

P	P	P	P	P	P
L	L	L	L	L	L
1	2	3	4	5	6

- Apply high-flow [Oxygen](#) to both the tracheostomy and the face
 - Remove speaking cap, if present
 - BVM – connect to tracheostomy with inner cannula, or pediatric mask if unable
 - If no chest rise, block tracheostomy with wet gauze and BVM on face
 - SpO₂ monitor
 - Look for easily correctible causes of distress, such as detached oxygen sources
 - If ventilator problem, remove patient from ventilator and utilize BVM
 - If no pulse at any time, CPR
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- [ETCO₂ monitor](#)
 - [12-lead ECG placement and acquisition](#)
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- If suction catheter cannot be passed, ventilate from mouth with standard airway management while covering stoma
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- Tracheal suctioning
 - 2-3mL saline may be used to loosen secretions
 - [IV](#) / [IO](#) access as appropriate for patient condition
 - If bleeding profusely, place 20cc of air or saline into cuff
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- If tube is blocked, deflate cuff if present and remove/clean or attempt to place new one
 - Attempt oral intubation
 - Replace stoma with ETT (one size below tracheostomy size or 6.0)
 - Direct the placement of digital pressure on the posterior wall of the manubrium
 - If this requires removal of the tracheostomy tube, only perform after successful intubation (will need to go deeper)

Consult Online Medical Control As Needed

Call OLMC before attempting to replace a new (<7 day) tracheostomy



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Pearls:

- Work with the family and other primary caregivers to troubleshoot any medical equipment, they are the best source of understanding
- Provide supplemental oxygen early through both the tracheostomy/stoma and face
- Call for assistance early
- A short bleed followed by a pause and then more severe bleeding is concerning for innominate bleed
- Remember that the respiratory issue may not be related to the tracheostomy; these patients still suffer from Asthma, COPD, CHF, etc.
- Consider surgical airways only when all other methods of oxygenation, ventilation, and securing the airway have failed
- Pediatric ETTs are sized according to age:
 - Premie: 2.5mm
 - 0-3mo: 3.0
 - 3-7mo: 3.5
 - 7-15mo: 4.0
 - 15-24mo: 4.5
 - 2-15yrs: $(\text{Age}+16)/4$
- Posterior manubrium pressure:

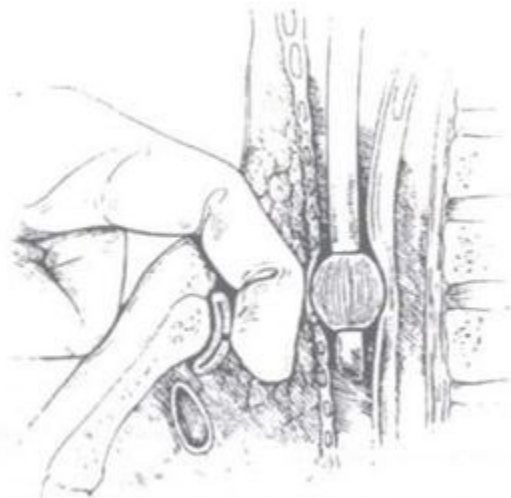


Figure 2: Manual compression of fistula